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Mini Project Proposal - BAI506

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STUDENT PLACEMENT PREDICTION AND READINESS ANALYSIS USING RESUME INTELLIGENCE AND MACHINE LEARNING		
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INITIAL PROJECT PROPOSAL

Campus placement is an important outcome for students as well as a key indicator for academic institutions. This mini project proposes a machine-learning-based system that predicts a student's placement likelihood from academic, technical and employability-related attributes and also provides a simple readiness analysis. The work will use the open-source campus-placement dataset as the initial training source, supplemented by a controlled synthetic dataset for development and robustness experiments. Data contributed by students, colleges and companies will initially be maintained as an independent external evaluation dataset and, after validation and availability of actual placement outcomes, suitable records may be incorporated into future training cycles. The proposed implementation will introduce a leakage-aware preprocessing pipeline by removing identifiers and post-placement information such as salary from prediction inputs, encoding categorical variables, scaling features when required, and using stratified train-test splitting with cross-validation. Additional features such as coding-platform activity, projects, internships, certifications and technical skills will be considered, including the number of problems solved, rank and rating/score on platforms such as LeetCode and HackerRank where the information is available. A document-processing module will also use OCR and NLP-based information extraction to extract relevant fields from resumes and placement forms, followed by validation and user confirmation before prediction. Logistic Regression, Decision Tree, Random Forest and Support Vector Machine classifiers will be trained and compared using accuracy, precision, recall, F1 score and confusion-matrix analysis. The selected model will be integrated into a Streamlit-based interface in which users can enter a student profile or upload a resume/form and obtain a Placed/Not Placed prediction, an estimated placement probability or readiness level, and the factors that most strongly influenced the result. The system will also provide basic improvement suggestions based on weaker academic, technical or employability indicators. The expected outcome is an accessible decision-support prototype that supports real-world external evaluation, progressive dataset expansion and future model retraining while keeping the final recruitment decision with human stakeholders.

CHAPTER 1: INTRODUCTION

Campus placement is a major transition point between higher education and employment. A student's placement outcome is influenced by several academic, technical and employability-related factors, including secondary and higher-secondary performance, degree percentage, employability-test score, specialization, prior work experience, internships, projects, certifications and demonstrated coding activity. Because many of these attributes are available before or during the placement season, they can be analysed using supervised Machine Learning (ML) to discover patterns associated with previous placement outcomes and to estimate the likelihood of future placement.

The selected open-source Student Placement Prediction project provides a suitable baseline for this mini project. Its documented workflow includes data cleaning, Exploratory Data Analysis (EDA), feature engineering, model training, model evaluation and final placement prediction. The baseline campus-placement dataset contains academic percentages, board and stream information, degree type, work experience, employability-test score, MBA specialization and placement status. Salary is present only for placed students and therefore must not be treated as a pre-placement predictor. The proposed work will retain this dataset as the initial training source while extending the feature space and establishing a separate real-world evaluation stream.

Our project will retain the core placement-prediction objective while modifying the baseline into a more complete decision-support prototype. The planned additions are leakage-aware preprocessing, stratified validation, comparison across several evaluation measures, placement probability/readiness output, feature-level explanations, coding and employability indicators, OCR/NLP-based resume and form extraction, and a Streamlit-based interface. The project will also maintain separate internal and external evaluation datasets so that data supplied by students, colleges and companies can be used to assess generalization beyond the original Kaggle dataset. After the actual placement outcome becomes available and the record passes data-quality and leakage checks, validated records may be incorporated into future retraining datasets. The system is intended to support students and placement coordinators by highlighting patterns and areas that may require attention; it is not intended to replace interviews, recruiter judgement or institutional placement policy.

1.1 Motivation

Students often receive useful feedback only after aptitude tests, interviews or recruitment drives have already taken place. A data-driven readiness estimate can provide earlier guidance by showing whether a student profile resembles historically placed or non-placed profiles. Placement cells can also use aggregated model outputs to identify groups that may benefit from additional aptitude practice, academic support or employability training. The motivation is therefore to convert historical placement records into an understandable, repeatable and accessible tool that supports early intervention while keeping final decisions with human stakeholders.

A further motivation is to reduce manual data-entry effort and make the system compatible with the way student information is commonly stored. Resumes and placement forms contain useful structured and semi-structured information about education, skills, projects, internships,

certifications and coding activities. An OCR and NLP pipeline can extract these attributes into a structured candidate profile, after which the student can verify or correct the extracted values before prediction. When actual placement outcomes are later known, validated records can contribute to external evaluation and future model improvement.

CHAPTER 2: LITERATURE SURVEY

The literature survey reviewed 17 referenced studies covering classical machine learning, ensemble learning, feature selection, coding and employability indicators, deep learning, representation learning, explainability and OCR.

Sl. No.	Authors & Year	Paper Title	Study / Methodology	Advantages	Limitations
1	P. Chouhan, S. Borkar and S. Gupta (2026)	Predicting Student Placement Readiness: A Machine Learning Approach Using Coding Activities and Multi-Dimensional Performance Indicators	Proposed ML-based placement-readiness prediction using academic performance, coding activity, aptitude and behavioural/co-curricular indicators. Used Logistic Regression, Random Forest, Gradient Boosting, SVM and XGBoost with SHAP interpretation.	Uses multidimensional student indicators and provides interpretable predictions using SHAP. Compares multiple ML models.	Uses a broader set of coding and behavioural indicators, making the system more complex. May require more detailed student data.
2	R. R. Velamala et al. (2026)	Comparative Analysis of Classical Machine Learning Models and a Variational Quantum Classifier for Student Placement Prediction	Compared Logistic Regression, SVM, Random Forest, MLP, XGBoost and a Variational Quantum Classifier using leakage-free preprocessing and cross-validation.	Rigorous validation and leakage-free preprocessing improve reliability. Provides extensive model comparison.	Focuses heavily on classifier benchmarking, including quantum methods, rather than student-oriented readiness feedback.
3	B. Bhujade, A. Khy & S. Dholay (2026)	AI-Driven Employability Prediction: Integrating Machine Learning and Educational Analytics for Student Placements	Proposed an AI/ML-based employability prediction framework using SVM, Random Forest, Gradient Boosting and XGBoost with academic performance, technical skills, internships and leadership experience.	Demonstrates the use of multiple student attributes for predicting employability and potential job roles.	Uses a broader employability profile, while the proposed project focuses on a compact campus-placement dataset and readiness-oriented interpretation.
4	K. Eswara Rao et al. (2025)	Student Placement Performance Analysis and Prediction Systems at Higher Education Institutions in South India Region: A Machine Learning Approach with Ranked Features	Analysed student placement performance using 39 features and feature-selection methods including Information Gain, Gain Ratio, Gini Index and Chi-square.	Identifies influential academic, attendance and coding-related factors. Uses systematic feature-selection techniques.	Uses a relatively large synthetic feature set and extensive feature-ranking methods. May be less practical for a simple student-facing system.
5	A. H. Elias, F. A. Khairi and A. H. Elias (2025)	Hybrid Machine-Learning Framework for Predicting Student Placement	Developed a hybrid stacking ensemble using SVM, Random Forest and Logistic Regression on student placement data.	Ensemble learning can combine the strengths of multiple classifiers. Identifies important predictors such as CGPA and previous academic performance.	Primarily focuses on classification and ensemble performance. Does not emphasize personalized readiness analysis or improvement guidance.
6	A. D. Praveen Kumar et al. (2025)	Predicting Student Employability Using Machine Learning: A Comparative Study of Classification Algorithms	Compared Logistic Regression, Random Forest and XGBoost using academic, technical, project, internship, certificate, extracurricular and aptitude attributes.	Considers a broad student profile and compares several effective ML algorithms. Includes technical and extracurricular factors.	Requires many student attributes and uses a broader application architecture. May involve greater data-collection complexity.
7	A. Pathak et al. (2025)	A Machine Learning Framework for Predicting	Developed an ML framework using preprocessing, EDA,	Provides a systematic ML pipeline with proper	Mainly concentrates on prediction and model

Student Placement Prediction and Readiness Analysis Using Machine Learning

Sl. No.	Authors & Year	Paper Title	Study / Methodology	Advantages	Limitations
		Student Placement Outcomes	categorical encoding, scaling, hyperparameter optimization and cross-validation.	preprocessing and model optimization.	development rather than providing student-specific readiness feedback.
8	Raghavendra C. K. et al. (2025)	Identifying Patterns and Trends in Campus Placement Data Using Machine Learning	Investigated patterns and trends in campus placement using multiple ML algorithms and ensemble techniques with skill-related data.	Helps identify placement-related patterns and important employability factors. Supports comparison of multiple ML approaches.	Focuses mainly on pattern discovery and model performance. Limited emphasis on actionable, individualized student guidance.
9	M. Ruparel & P. Swaminarayan (2025)	Enhanced Student Placement Prediction Using Machine Learning: A Comparative Evaluation of Algorithms	Compared Logistic Regression, Random Forest, Decision Tree, Naïve Bayes, SVM, KNN, Gradient Boosting and LDA for placement prediction.	Shows the importance of algorithm comparison, preprocessing, feature scaling and outlier handling in placement prediction.	Supports the proposed multi-model comparison using Logistic Regression, Decision Tree, Random Forest and SVM.
10	Anonymous/IEEE conference authors (2025)	A Review of Various Deep Learning Approaches for Placement Prediction	Reviewed deep-learning approaches for placement prediction, focusing on self-attention, feature importance, class imbalance and ensemble methods.	Highlights the potential of deep learning for complex student-placement datasets while emphasizing interpretability and evaluation metrics.	The proposed project deliberately uses lightweight classical ML models suitable for a mini-project and smaller dataset.
11	S. Abuthahir et al. (2025)	Machine Learning-Based Student Placement Prediction Analysis: A Data-Driven Approach to Enhance Employability	Used academic performance, extracurricular activities, certifications, aptitude scores and soft-skill ratings for ML-based placement prediction.	Demonstrates the usefulness of combining academic and non-academic attributes for placement prediction.	Supports extending placement analysis beyond academic percentages while maintaining a pre-placement feature set.
12	Employability Prediction Study (2025)	Leveraging LLMs for Optimised Feature Selection and Embedding in Structured Data: A Case Study on Graduate Employment Classification	Investigated LLM/BERT-based embeddings and feature selection for graduate employment classification using ANN, CatBoost and BERT-based approaches.	Feature selection and BERT-based representations improved classification performance in the studied graduate-employment dataset.	Shows a recent direction toward advanced representation learning; the proposed project uses conventional interpretable models for a simpler deployment architecture.
13	V. S. Agrawal and S. S. Kadam (2024)	Predictive Analysis of Campus Placement of Student Using Machine Learning Algorithms	Applied Logistic Regression, Decision Tree, Random Forest, SVM, XGBoost, AdaBoost, KNN and Naive Bayes to historical placement records.	Provides extensive comparison of classification algorithms and feature-importance analysis.	Primarily focuses on historical placement prediction. Does not emphasize readiness analysis, leakage-aware inputs or an interactive student interface.

2.1 Limitations of Existing System

- The commonly used campus-placement datasets are collected from specific institutions and may not accurately represent all colleges, disciplines, or recruitment environments.
- The inclusion of post-placement attributes such as salary can introduce target leakage, leading to unrealistic prediction performance.
- Many existing studies rely heavily on accuracy as the primary evaluation metric, which may not adequately reflect model performance across different classes.
- Several placement prediction systems provide only a binary *Placed/Not Placed* result without explaining the factors influencing the prediction.
- Most implementations are limited to research notebooks and lack user-friendly interfaces for students and placement coordinators.
- Many existing systems depend on manually entered structured data and do not provide OCR/NLP-based extraction from resumes or placement forms.
- Most studies evaluate models on a single historical dataset and do not clearly separate independent real-world external evaluation from future training data expansion.

2.2 Outcome of the Literature Survey

The literature survey indicates that machine learning techniques are effective for predicting student placement and employability outcomes. Existing studies highlight the importance of data preprocessing, feature selection, model comparison, and comprehensive evaluation using metrics such as Accuracy, Precision, Recall, and F1-Score. Recent work also considers coding activity, behavioural indicators, synthetic data and advanced representation learning. However, there remains a need for a practical and user-friendly system that combines reliable prediction, model interpretability, placement-readiness analysis, document-based feature extraction and independent real-world evaluation.

The proposed project addresses these limitations by utilizing only appropriate pre-placement attributes, implementing a reusable preprocessing pipeline, comparing multiple classification algorithms, and selecting a model using balanced evaluation metrics. The final system will provide placement predictions through an interactive web interface, accept structured input or resume/form documents, extract relevant information using OCR and NLP, provide readiness-oriented insights and improvement suggestions, and evaluate generalization using independently collected student, college and company data. Validated records with known outcomes can subsequently support controlled dataset expansion and model retraining.

CHAPTER 3: PROBLEM STATEMENT

Placement preparation is usually based on a mixture of academic performance, aptitude preparation, communication skills, technical skills, coding practice, projects, internships and recruiter-specific requirements. Students may find it difficult to interpret these factors together, while placement coordinators may have many student records and resumes to review. There is therefore a need for an accessible system that can analyse available pre-placement academic and employability attributes, extract relevant information from resumes and forms, and estimate whether a student profile is likely to be associated with a placed or not-placed outcome.

The technical problem is to construct a reproducible supervised-learning pipeline that avoids leakage from identifiers and post-placement fields, processes numerical, categorical and document-derived variables consistently, compares suitable classification algorithms, and chooses a model using more than one evaluation measure. The pipeline must distinguish the initial Kaggle training data from independently collected external evaluation data, support synthetic-data experiments, and provide a controlled mechanism for incorporating validated records into future training. The application should present the prediction in a form that can be understood by non-technical users, without presenting the output as a guarantee of employment.

3.1 OBJECTIVES

1. Model Development and Evaluation: To develop a leakage-aware Machine Learning pipeline using Logistic Regression, Decision Tree, Random Forest and Support Vector Machine models, and compare them using stratified validation, accuracy, precision, recall, F1 score and confusion-matrix analysis.

2. User-facing Placement Readiness Interface: To build a Streamlit-based interface that accepts structured student attributes and uploaded resumes/forms, validates extracted information, and displays placement prediction, readiness information and important contributing factors.

3. Comparative Performance Analysis: To evaluate and compare the performance of different machine learning algorithms for student placement prediction and identify the most suitable model for deployment based on multiple evaluation metrics and external validation.

4. Document-Based Feature Extraction: To develop an OCR and NLP-based module that extracts relevant academic, technical, coding, project, internship and certification information from resumes and placement forms for structured model input.

5. Feature Expansion: To incorporate additional pre-placement indicators such as LeetCode/HackerRank problems solved, rank and rating/score where available, along with programming skills, projects, internships and certifications.

6. External Validation and Dataset Expansion: To use data contributed by students, colleges and companies as an independent external evaluation dataset and, after verified outcomes and quality checks, establish a controlled pathway for future model retraining.

7. Synthetic Data Experimentation: To use the available high-volume synthetic dataset for controlled development, robustness and feature-ablation experiments without treating synthetic records as a substitute for real-world external evaluation data.

3.2 NOVELTY OF THE PROPOSED SYSTEM

The novelty of the proposed system lies in combining leakage-aware student placement prediction with placement-readiness analysis, interpretable decision support, document-based feature extraction and real-world external validation in a single user-facing system. Unlike a basic placement classifier that produces only a Placed/Not Placed output, the proposed system follows a structured preprocessing pipeline that excludes identifiers and post-placement attributes, applies consistent feature transformation, and uses stratified validation for reliable evaluation.

The system further extends placement prediction by providing an estimated placement probability or readiness level, identifying important factors associated with the prediction, and generating simple improvement-oriented suggestions. Resume and form documents can be processed through OCR and NLP to reduce manual data entry and derive structured features such as education, technical skills, projects, internships, certifications and coding-platform indicators. The system also distinguishes internal model evaluation from external evaluation using student, college and company data, creating a controlled pathway for validated records to support future retraining.

Recent research has also explored placement prediction, feature selection, ensemble learning, readiness analysis, coding activity and synthetic data separately. The proposed project focuses on integrating a simpler, reproducible and educationally oriented workflow around pre-placement attributes, leakage control, OCR/NLP-assisted feature extraction, coding-platform indicators, external validation and progressive dataset expansion.

3.3 SCOPE AND ASSUMPTIONS

- The prototype will use the publicly available Kaggle campus-placement dataset as the initial training data. The target variable is placement status: Placed or Not Placed. A separate high-volume synthetic dataset of approximately 5,000–10,000 records will be used for controlled development, robustness and feature experiments where appropriate.
- Serial number and salary will not be used as predictive inputs. Salary is an outcome available only after placement and would otherwise leak target information into the model. Similarly, any feature that becomes known only after or because of a recruitment outcome will be excluded from pre-placement prediction.
- The system will be treated as an educational decision-support tool. It will not rank students for recruiters, make admission decisions, or replace human evaluation during campus recruitment. Resume/form extraction will include a verification step so that OCR or NLP errors can be corrected before the prediction is generated.
- The final interface and model pipeline will be structured so that data contributed by students, colleges and companies can first be used for independent external evaluation. After actual placement outcomes are available, records will undergo validation, duplicate checking, privacy checks and leakage checks before suitable records are considered for future retraining. The system will retain source and collection metadata to support dataset traceability and model-version analysis.

CHAPTER 4: PROPOSED METHODOLOGY

The project will follow the end-to-end workflow shown in Figure 1. The Kaggle campus-placement dataset will serve as the initial training source, while synthetic data will support controlled experimentation and robustness testing. Data contributed by students, colleges and companies will initially remain separate for independent external evaluation. Resume and form documents will be processed using OCR and NLP to extract validated structured features. Records with verified outcomes may subsequently contribute to future model retraining.

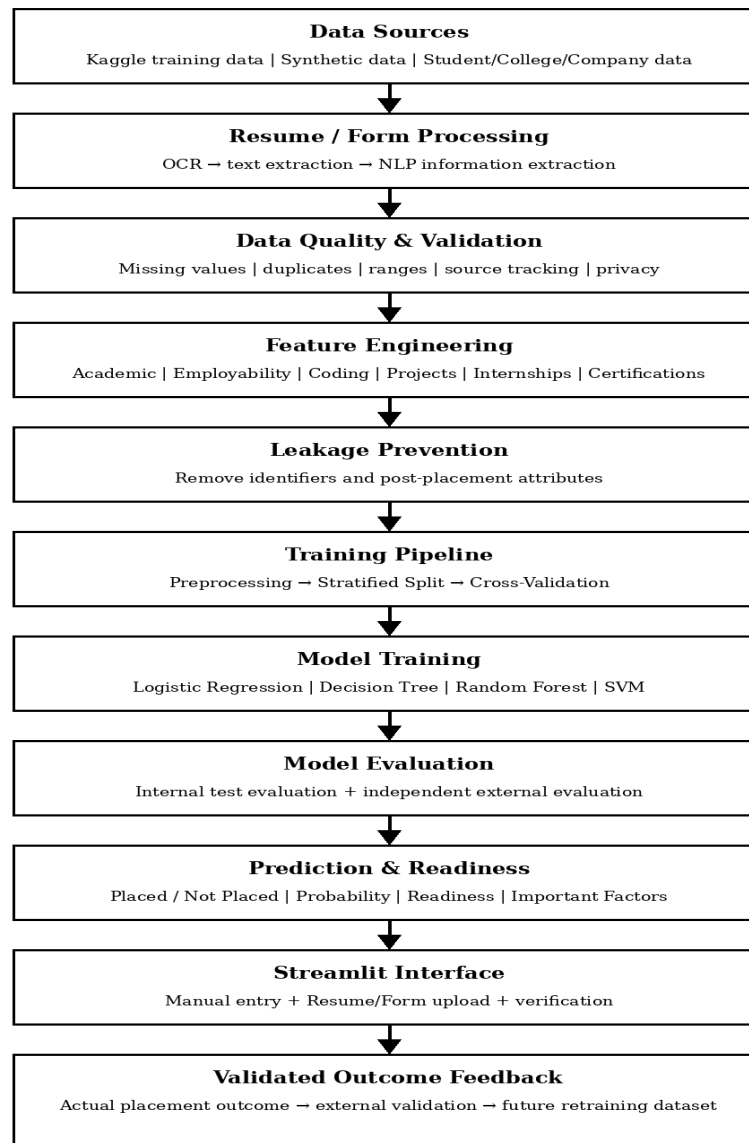


Figure 1. Proposed workflow for student placement prediction and readiness analysis.

1. Data Sources and Dataset Strategy:

The system uses three distinct data streams: (a) the Kaggle campus-placement dataset for initial model training, (b) the available synthetic dataset for controlled development and robustness experiments, and (c) real-world data contributed by students, colleges and companies for independent external evaluation. The third source will be considered for future training only after the actual placement outcome is known and the record passes validation, privacy, duplicate and leakage checks.

Working: Each record is mapped to a common feature schema, while data-source and collection metadata are retained for traceability. The placement status (Placed/Not Placed) is the target when an actual outcome is available.

2. Resume/Form OCR and Information Extraction:

Students or coordinators can upload resumes or placement forms as supported PDF/image documents. OCR converts document images into machine-readable text, after which NLP-based section detection and information extraction identify education, technical skills, projects, internships, certifications, coding-platform indicators and other relevant pre-placement attributes. **Working:** Extracted values are normalized into the project's structured feature schema. Low-confidence or ambiguous fields are flagged and shown for user verification before they are passed to the prediction pipeline.

The system uses a historical campus placement dataset containing academic and employability attributes such as SSC, HSC, degree percentage, work experience, employability test score, and specialization. The placement status (Placed/Not Placed) is the target variable.

Working: The dataset is loaded into Python using Pandas and used for model development.

3. Data Cleaning & Pre-processing:

The collected data is prepared for machine learning by handling missing values, categorical attributes, irrelevant/inconsistent data and validated document-derived fields.

Working: Categorical features are encoded into a suitable format, input features are separated from the target variable, and document/platform features are checked against valid ranges and category definitions.

4. Exploratory Data Analysis (EDA):

EDA is performed to identify important patterns, distributions, correlations, class balance and differences between training and external evaluation sources.

Working: Visualizations and statistical analysis are used to understand the relationship between student attributes and placement outcomes.

5. Stratified Split & Cross-Validation:

The prepared Kaggle training dataset is divided into training and testing sets while maintaining the proportion of Placed and Not Placed students. Independently collected real-world data remains outside the training split for external evaluation.

Working: Stratified splitting and cross-validation are used to obtain reliable internal evaluation, while the external dataset is evaluated only after the model pipeline has been finalized for the relevant experiment.

6. Model Training:

Four classification algorithms are trained on the prepared training data:

- LogisticRegression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)

Working: The models learn patterns between academic, employability, technical and other validated pre-placement attributes and placement status. Feature-ablation experiments can compare the baseline feature set with the expanded feature set.

7. Model Evaluation & Selection:

The trained models are tested and compared using internal validation metrics and, after model selection for an experiment, independent external evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Working: Evaluation results are compared across models and across internal versus external datasets to understand predictive performance and generalization.

8. Prediction & Readiness Analysis:

The selected model is used to predict the placement outcome for a student's manually entered or document-extracted input data.

Working: The system generates Placed/Not Placed predictions, prediction probability where applicable, identifies important factors related to the prediction, and provides readiness-oriented insights. OCR/NLP extraction errors are not treated as model signals until the extracted values are validated.

9. Streamlit Interface:

A Streamlit-based interface is provided for students and placement coordinators, with support for structured input and resume/form upload.

Working: Users enter student information or upload a supported resume/form. Extracted or manually entered information is validated, processed through the same preprocessing pipeline and passed to the trained model. The prediction, readiness information and important factors are then displayed.

4.1 Dataset and Input Feature Plan

The initial training source is the Kaggle campus recruitment dataset containing academic, background and employability attributes. The project will extend the input schema with validated technical, coding, project, internship and certification features when these are available in external records or documents. Table 1 summarizes the main inputs planned for the predictive pipeline. Categorical fields will be encoded and numerical fields will pass through the appropriate preprocessing steps. The target is placement status.

The proposed system uses the Factors Affecting Campus Placement dataset, stored locally as Placement_Data_Full_Class.csv, as the initial training dataset. The dataset contains 215 student records with academic and employability attributes such as SSC percentage, HSC percentage, degree percentage, degree type, work experience, employability test score, MBA specialization, MBA percentage, and placement status. In addition, the project has a separate synthetic dataset of approximately 5,000–10,000 records for controlled experimentation. Real-world records contributed by students, colleges and companies will initially be reserved for external evaluation and can later be added to the retraining dataset after outcome verification and quality checks.

The Kaggle dataset is integrated into the machine learning pipeline using Pandas and Scikit-learn.

The status attribute is used as the target variable (Placed or Not Placed), while appropriate pre-placement attributes serve as input features. Resume/form documents are processed separately through OCR and NLP before their extracted attributes are mapped to the same feature schema. Four classification algorithms—Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM)—are trained and evaluated. Their performance is compared using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix for both internal and external evaluation where the ground-truth placement outcome is available.

Feature Group	Examples from Dataset / New Inputs	Role in Proposed Model
School performance	ssc_p, hsc_p	Numerical academic inputs
School background	ssc_b, hsc_b, hsc_s	Categorical background inputs
Degree performance	degree_p, degree_t, CGPA	Academic percentage and degree type
Work experience	workex, internship duration	Pre-placement experience indicators
Employability	etest_p, aptitude/assessment score	Employability-related inputs
MBA attributes	specialisation, mba_p	Specialization and percentage
Coding activity	leetcode_problems_solved, hackerrank_problems_solved	Technical/problem-solving indicators
Coding performance	leetcode_rank, leetcode_rating, hackerrank_rank, hackerrank_score	Platform performance indicators where available
Technical skills	programming languages, frameworks, tools	Structured technical-skill inputs extracted from forms/resumes
Projects	project_count, project domains, technologies	Project-based employability indicators
Certifications	certification_count, relevant certification categories	Skill-validation indicators
Document-derived data	Education, skills, projects, internships, certifications extracted using OCR/NLP	Automatically extracted candidate features after validation
Excluded fields	sl_no, salary, identifiers and post-placement attributes	Identifier / target-leakage exclusion
Target	status	Placed or Not Placed

Table 1. Main dataset fields and their planned use

4.1.1 External Evaluation and Data Integration

Data contributed by students, colleges and companies will initially be stored separately from the Kaggle training data and used as an independent external evaluation dataset. The purpose is to determine whether a model trained on the baseline data generalizes to newer students, different institutions and different recruitment environments. External records will be mapped to the

common feature schema, checked for missing or inconsistent values, and evaluated only when the actual placement outcome is available.

After external evaluation, records with verified outcomes may be considered for future retraining. Before incorporation, duplicate records, invalid values, personally identifying information, temporal leakage and post-placement fields will be checked. The source and collection period will be retained as metadata so that model versions can be evaluated against different data cohorts.

4.1.2 Resume/Form OCR and Feature Extraction

The document-processing module accepts supported resume or form documents and applies OCR to convert image-based content into text. NLP-based extraction then identifies relevant entities and sections such as education, skills, projects, internships, certifications and coding-platform information. Extracted values are normalized and validated before being converted into model-ready features. Where a field is ambiguous or has low extraction confidence, the interface will request confirmation from the user rather than silently accepting the value.

The same feature schema is used for manual input and document-derived input so that the model receives consistent representations regardless of how the candidate data was collected.

4.2 Model Training and Evaluation

Data Split and Validation: A stratified train-test split is performed on the Kaggle-derived training data to maintain the proportion of placed and not placed students in both datasets. Cross-validation is applied to reduce dependence on a single random split. The real-world student, college and company dataset is kept independent for external evaluation and is not used to tune the model before the external test.

Model Training: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM) classifiers are trained using the preprocessed training dataset. Basic hyperparameter tuning may be performed using only the training/validation data. Baseline and extended-feature experiments can be conducted to measure the effect of coding, project, internship, certification and document-derived features.

Model Evaluation and Selection: The trained models are compared using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. Internal validation results are reported separately from external evaluation results. The model with balanced and stable performance is selected for the intended deployment experiment, while generalization differences between the two evaluation settings are documented.

4.3 Prediction, Interpretation and Interface

Prediction and Deployment: The selected model is saved and integrated with a Streamlit-based web interface. The interface accepts manual student information or a supported resume/form document. For uploaded documents, OCR and NLP extract candidate attributes, the user verifies the extracted values, and the same preprocessing pipeline is then applied. The system displays the predicted placement status, readiness level or probability, key factors influencing the prediction, and improvement-oriented suggestions.

CHAPTER 5: SYSTEM REQUIREMENTS SPECIFICATIONS (FUNCTIONAL AND NON- FUNCTIONAL)

5.1 Functional Requirements

- Accept academic, technical, coding and employability details through a user-friendly input form and support resume/form upload for document-assisted feature extraction.
- Validate numerical, categorical and OCR/NLP-extracted inputs before prediction, including user confirmation for ambiguous document-derived values.
- Apply the same preprocessing and feature-transformation steps used during model training to manual and document-derived inputs.
- Generate placement predictions, readiness analysis and important contributing factors.
- Display important factors influencing the prediction and basic improvement-oriented suggestions.
- Load the saved model for prediction without retraining and preserve separate external-evaluation data from the training pipeline.
- Store validated real-world records with source and outcome metadata so that suitable records can be evaluated externally and considered for future retraining.
- Extract relevant candidate information from supported resumes/forms using OCR and NLP and map the extracted information to the common feature schema.

5.2 Non-Functional Requirement

- **Usability:** The system should be easy to use for students and placement coordinators.
- **Reliability:** Consistent predictions should be generated for the same valid inputs.
- **Performance:** Predictions should be produced within a few seconds.
- **Maintainability:** The code structure should allow easy updates and modifications.
- **Privacy and Responsible Use:** The system should avoid unnecessary personal information and provide decision support rather than automated recruitment decisions.

- **Data Quality and Traceability:** Externally contributed and document-derived records should undergo validation, duplicate checking, leakage checks and source tracking before being used for evaluation or future retraining.

5.3 Software and Hardware Requirements

Requirement	Specification
Programming Language	Python 3.x
Libraries	Pandas, NumPy, Scikit-learn, Matplotlib/Seaborn, Joblib, OCR/NLP libraries
Development Tools	Jupyter Notebook and/or Visual Studio Code
Web Interface	Streamlit
Dataset	Placement_Data_Full_Class.csv, synthetic dataset, external evaluation data
Hardware	64-bit Laptop/Desktop, Minimum 4 GB RAM (8 GB Recommended)
Operating System	Windows, Linux, or macOS

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